

Calcium Channel Blockers



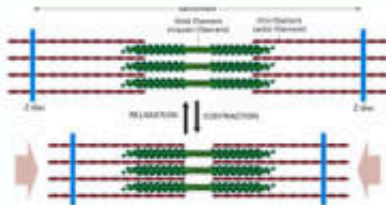
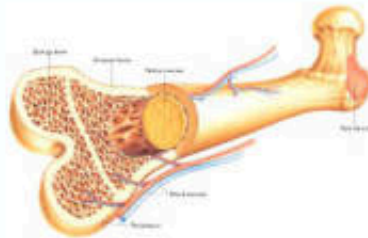
Krisada Sakchaisri, Ph.D.

วัตถุประสงค์

1. สามารถอธิบายกลไกการออกฤทธิ์และฤทธิ์ทางเภสัชวิทยาของยาต้านแคลเซียมกลุ่มต่างๆ ได้
2. สามารถบรรยายอาการไม่พึงประสงค์ อันเกี่ยวเนื่องจากฤทธิ์ทางเภสัชวิทยาของยาต้านแคลเซียมได้
3. สามารถอธิบายบทบาทและประโยชน์ทางการแพทย์ของยาต้านแคลเซียม

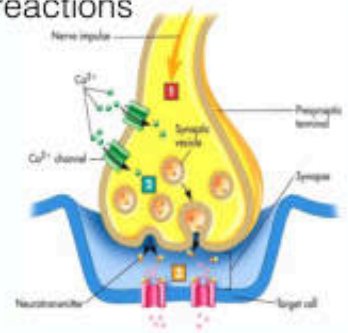
Functions of Calcium in The Body

- ✦ **Bone** development
- ✦ Teeth development

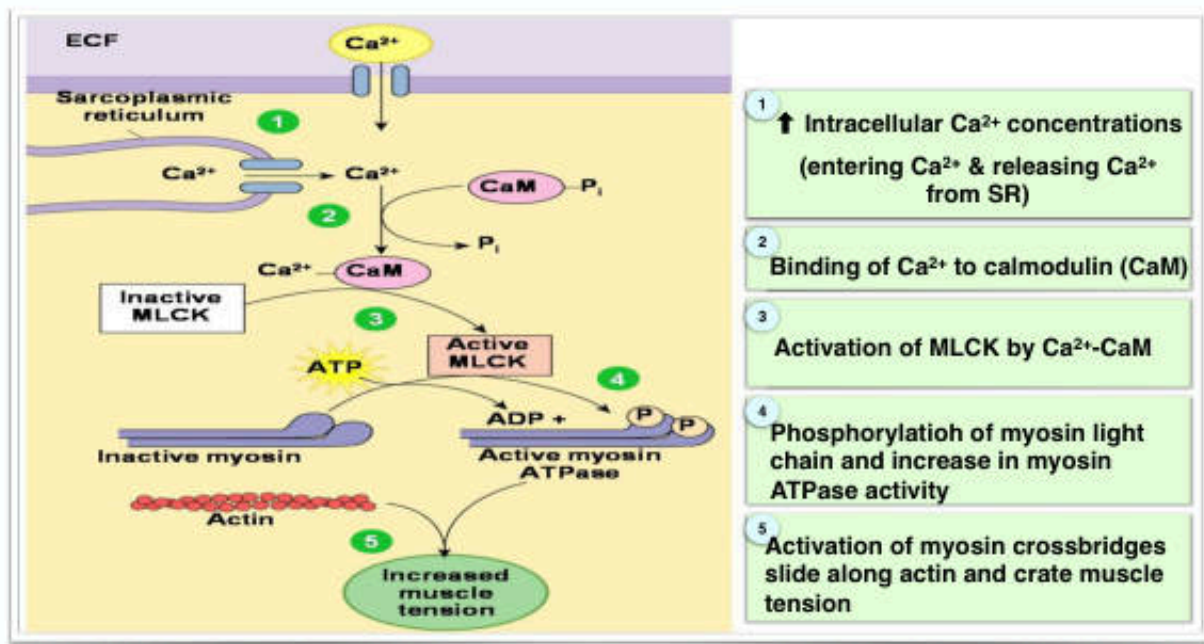


- ✦ **Contraction** and relaxation of muscle (including normal heart beat)
- ✦ Activation of enzyme reactions

- ✦ Transmission of nerve impulses
- ✦ Stimulation of hormone secretions

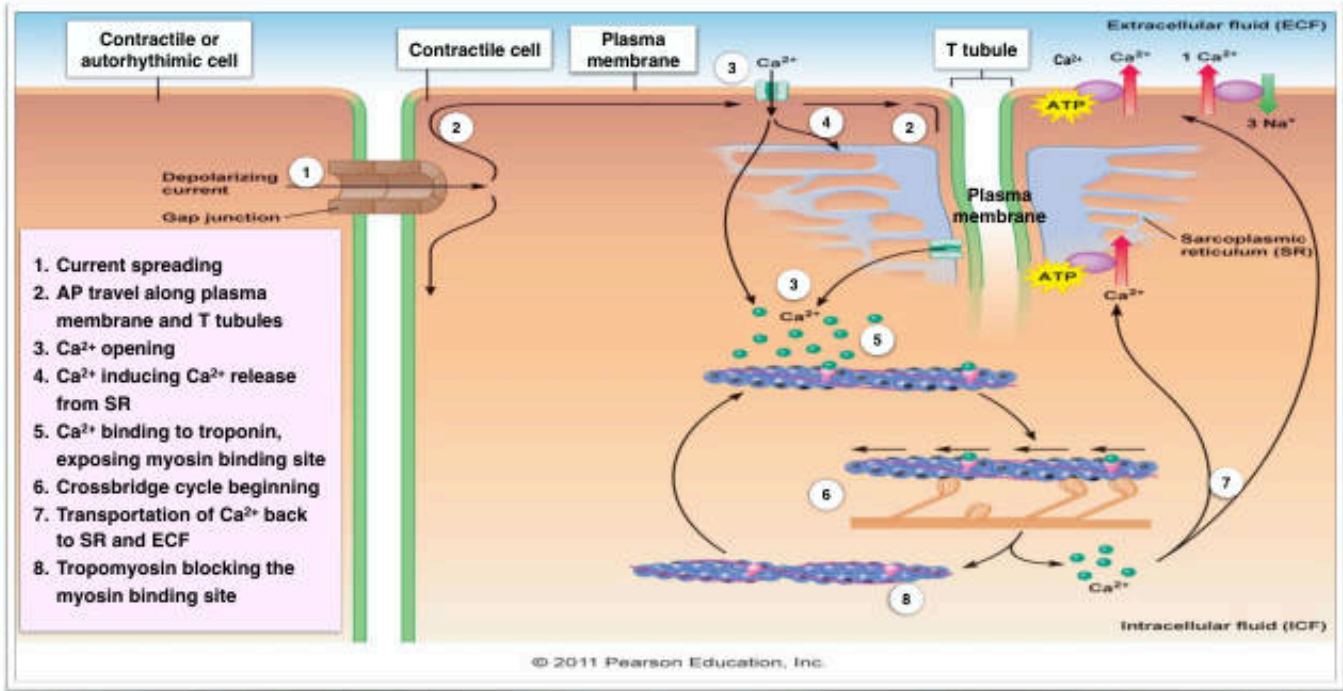


Smooth Muscle Contraction

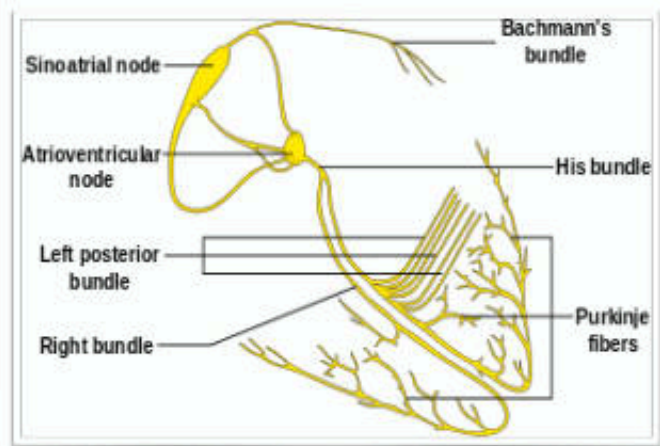
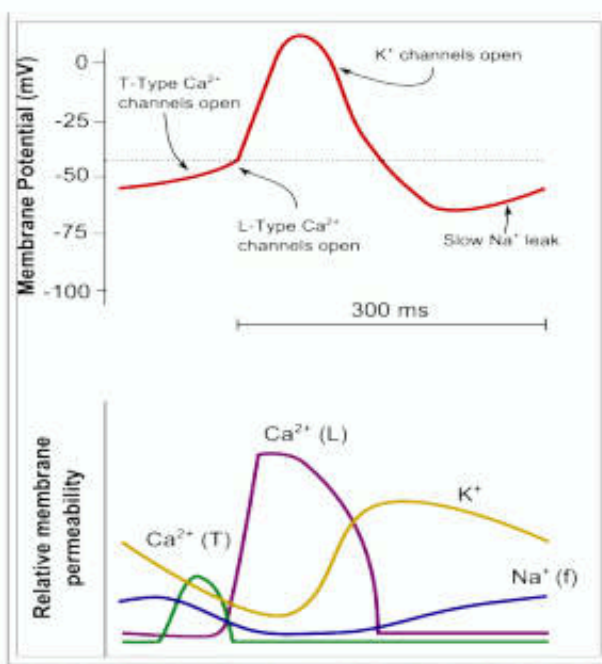


MLCK: Myosin light chain kinase

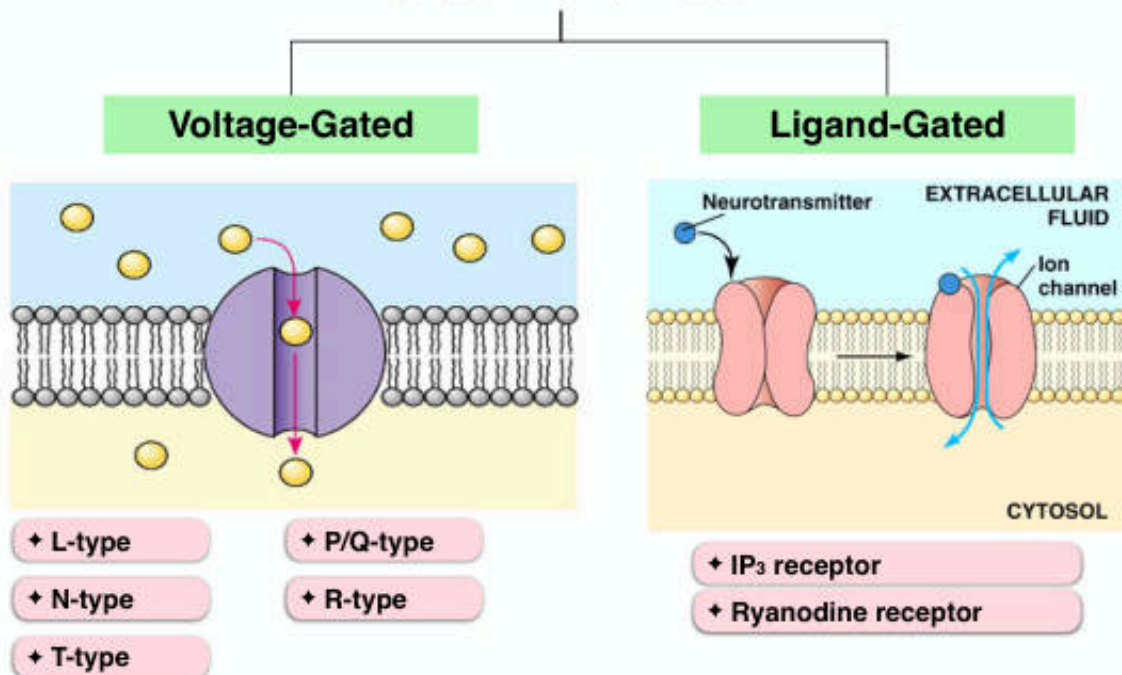
Cardiac Muscle Contraction



Pacemaker Action Potential

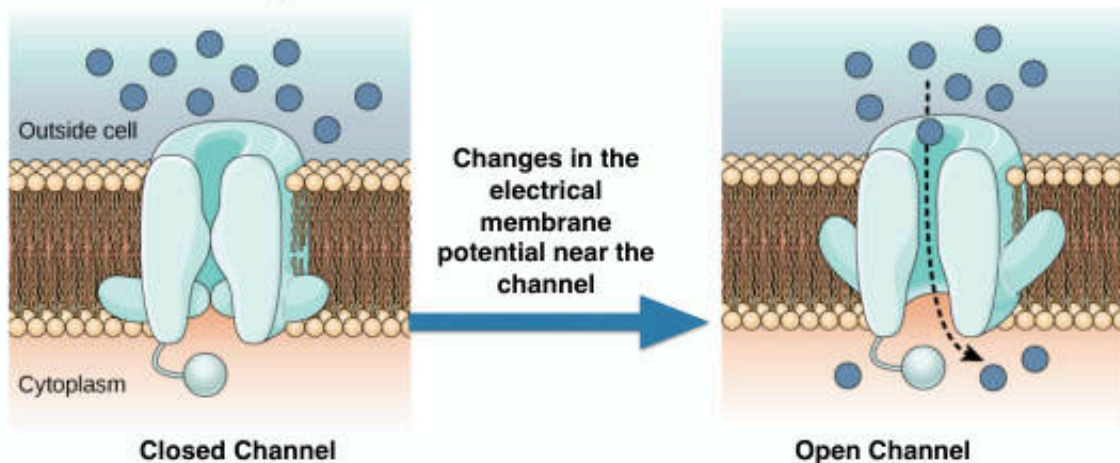


Calcium Channels



Ionotropic receptor: ligand-gated transmembrane ion channels

Voltage-Gated Calcium Channel



- ✦ Mediate calcium influx in response to membrane depolarization
- ✦ Regulate intracellular processes such as contraction, secretion, neurotransmission, and gene expression

Voltage-Gated Calcium Channel Subtypes

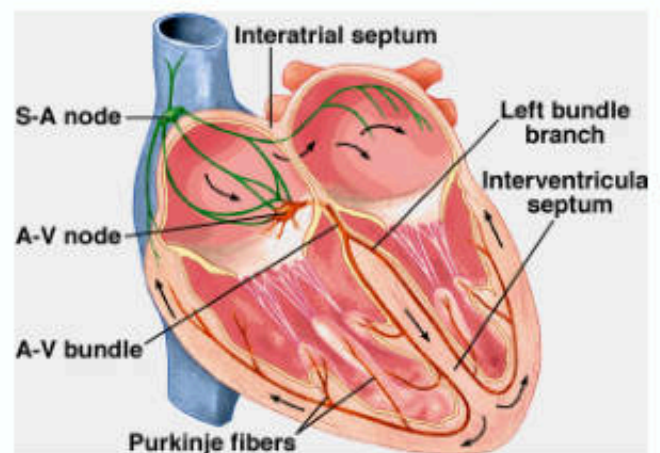
Type	Named for	Electrophysiology	Location	Cellular functions
L	Long-lasting	HVA, slow inactivation	Cardiac myocytes, smooth muscle, endocrine cells, neuronal cell bodies	Excitation-contraction coupling, hormone release, regulation of transcription, neurotransmitter release
T	Transient	LVA, rapid inactivation	Neuronal cell bodies and dendrites, cardiac and smooth muscle myocytes	Pacemaking, repetitive firing
N	Neuronal	HVA, moderate inactivation	Nerve terminals and dendrites, neuroendocrine cells	Neurotransmitter release, dendritic Ca^{2+} transients, hormone release
P/Q	Purkinje cell/ Q after P	HVA, moderate inactivation	Nerve terminals and dendrites, neuroendocrine cells	Neurotransmitter release, dendritic Ca^{2+} transients, hormone release
R	Remaining/ residual	HVA, fast inactivation	Neuronal cell bodies and dendrites	Repetitive firing, dendritic calcium transients

HVA: high voltage-activated; LVA: low voltage-activated

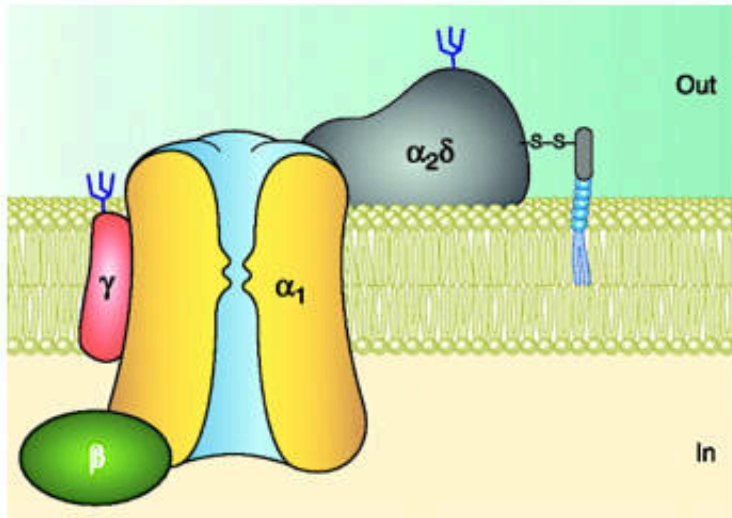
L-Type, Voltage-Gated Calcium Channels

In response to electrical depolarization, they mediate the entry of extracellular calcium into

- ❖ *Smooth muscle*
- ❖ *Cardiac myocytes*
- ❖ *Sinoatrial (SA) nodal cells*
- ❖ *Atrioventricular (AV) nodal cells*



Voltage-Gated Calcium Channels



Channel family	Current type	Contributing α_1 subunit
Ca_v1	L-type	$\alpha_{1.1}$, $\alpha_{1.2}$, $\alpha_{1.3}$, $\alpha_{1.4}$
Ca_v2	P/Q-type	$\alpha_{2.1}$
	N-type	$\alpha_{2.2}$
	R-type	$\alpha_{2.3}$
Ca_v3	T-type	$\alpha_{3.1}$, $\alpha_{3.2}$, $\alpha_{3.3}$

Auxiliary subunits:

β_1 , β_2 , β_3 , β_4

$\alpha_{2\delta 1}$, $\alpha_{2\delta 2}$, $\alpha_{2\delta 3}$, $\alpha_{2\delta 4}$

γ_1 , γ_2 , γ_3 , γ_4 , γ_5 , γ_6 , γ_7 , γ_8

Ca^{2+} channels are composed of α_1 , β , γ and $\alpha_2\delta$ subunits

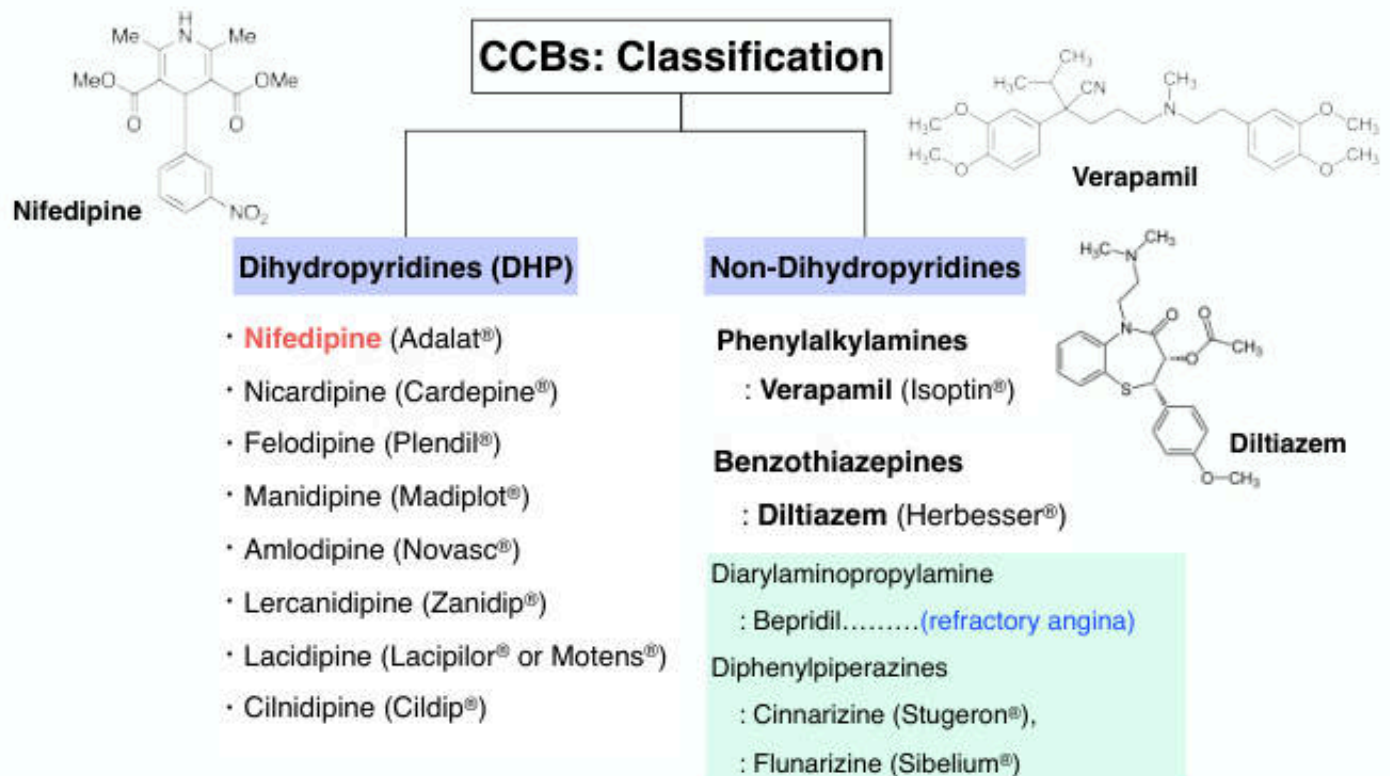
Physiol Rev 90: 1461–1506, 2010

Calcium Channel Blockers (CCBs)

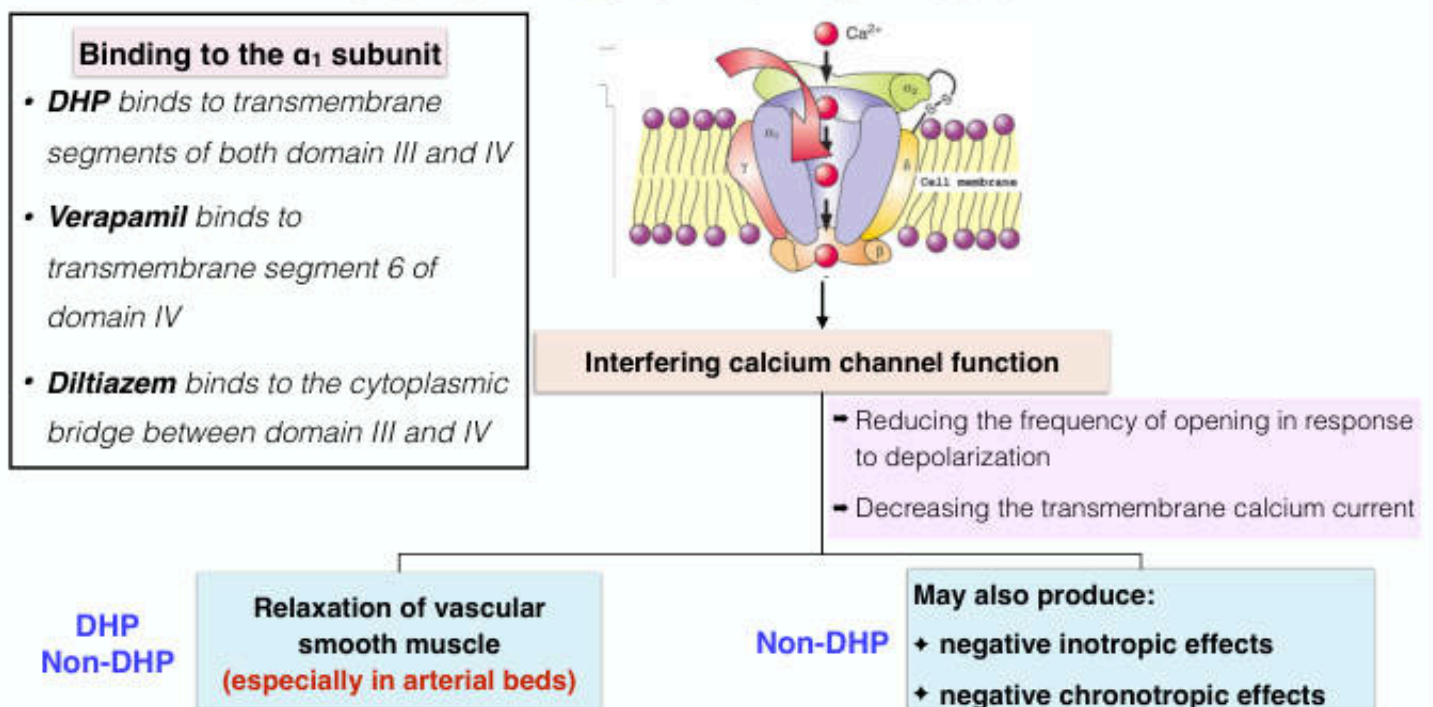
Calcium Antagonists

Calcium Channel Antagonists

Calcium Entry Blockers



CCBs: Mechanism of Action



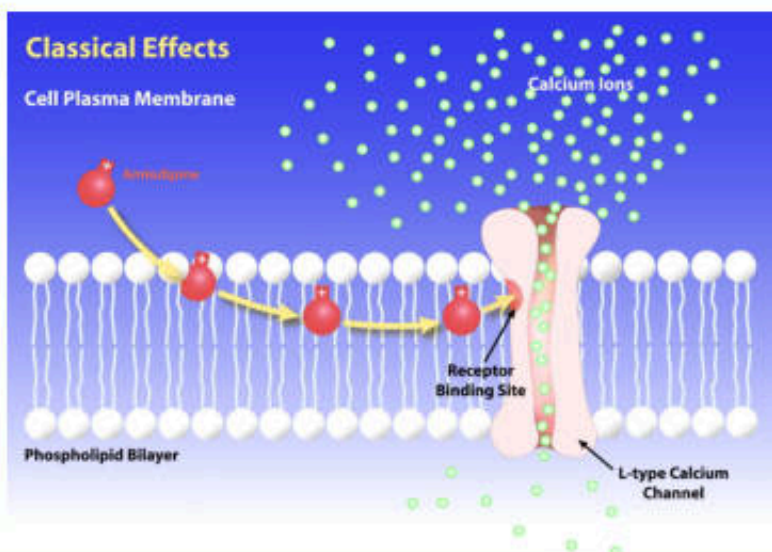
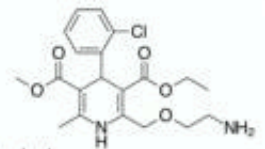
CCBs: Pharmacokinetics

	Nifedipine (immediate release)	Amlodipine	Verapamil	Diltiazem
Bioavailability	45 - 75	60 - 65%	25 - 35%	40
Protein binding	> 90	> 95	90	80 - 85
Metabolism	Hepatic (CYP3A4)	Hepatic (CYP3A4)	Hepatic (CYP3A4 and others)	Hepatic (CYP3A4)
Active metabolite	No	No	Yes (norverapamil)	Yes
Elimination half-life (h)	2 - 5	35 - 45	3 - 7 (slow-release~11)	4 - 6
Onset of action (min)	20	NA	30	30 - 60
Usual dosing frequency	TID - QID	Daily	TID - QID	TID - QID

NA: Not available

Adapted from *Drug Information Reference* 2003

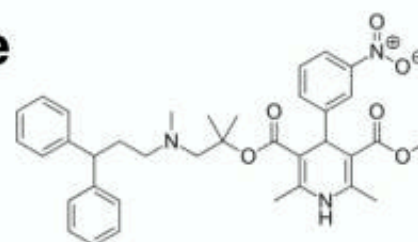
Amlodipine



- ★ It is in the charged state at physiologic pH
 - ⇒ strong electrostatic interactions with membrane phospholipid
- ★ Pleiotropic effects
 - ⇒ stimulating NO production
 - ⇒ acting as an antioxidant
 - ⇒ regulating membrane fluidity and cholesterol deposition

★ A sustained **reservoir** of amlodipine in the membrane at levels >10,000-fold higher than in the surrounding aqueous environment

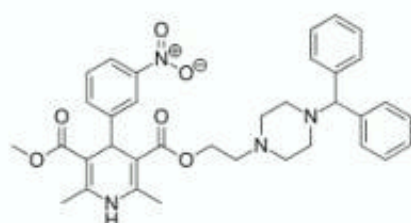
Lercanidipine



Additional Effects

- ★ Highly selective for vascular tissue
- ★ Highly lipophilic and is stored within cell membranes
 - ⇒ slow onset of action
 - ⇒ persistent smooth muscle relaxant effect
- ★ Short plasma half-life
- ★ Long duration of action
- ★ **Less peripheral edema**

- ★ Nephroprotective effect
 - ⇒ dilating both afferent and efferent arterioles
- ★ Antioxidant effects
- ★ Regression of left ventricular hypertrophy has also been reported



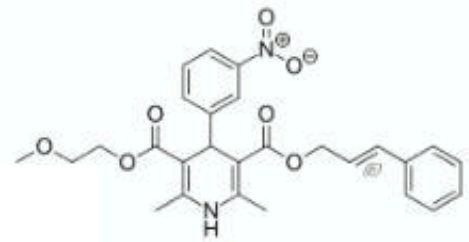
Manidipine: L/T-type CCB

T-type calcium channel

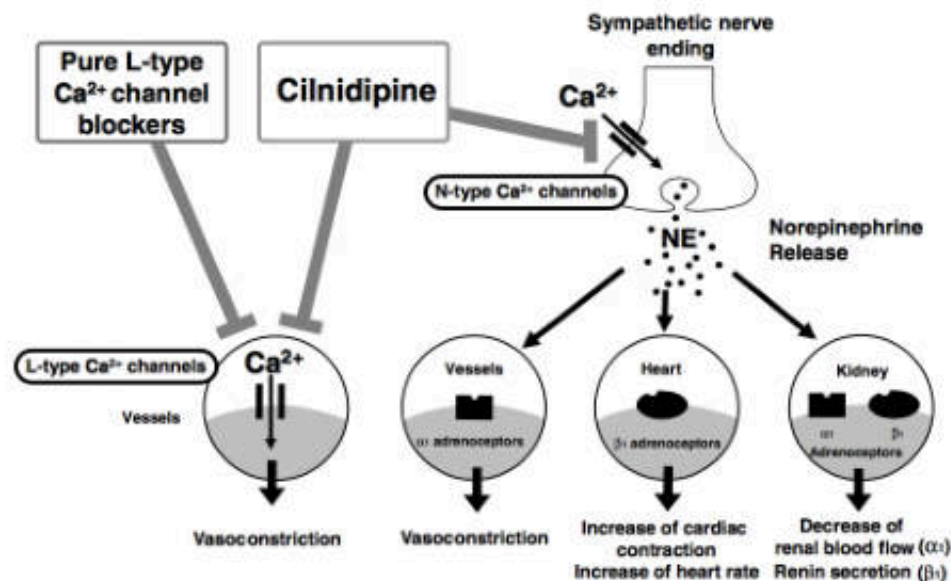
- ★ A growing body of evidence has accumulated depicting important roles of T-type Ca^{2+} channels in the regulation of cardiovascular function
 - ⇒ generation of pacemaker potential
 - ⇒ regulation of arterial resistance
- ★ T-type Ca^{2+} channels are also involved in cardiomyocyte growth and survival

- ★ The combined L-/T-type Ca^{2+} channel blockers produced a significantly greater inhibition of contraction than L-type Ca^{2+} channel blockers alone in small vessels
 - ⇒ a greater expression of T-type as Ca^{2+} channels in microvessels than L-type Ca^{2+} channels
- ★ It is believed that combined L-/T-type Ca^{2+} channel blockers equalize the hydrostatic pressure across the capillary bed through equal arteriolar and venular dilatation
 - ⇒ reducing vasodilatory edema

Cilnidipine: L/N-type CCB

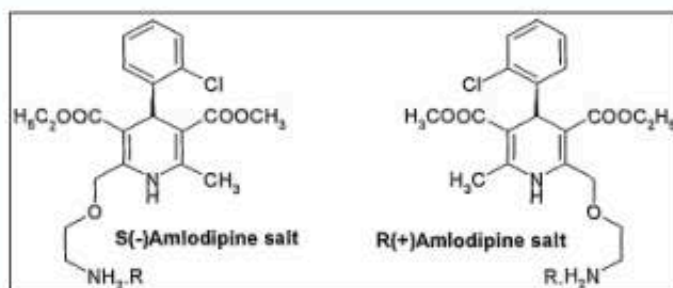


Cilnidipine is a unique Ca^{2+} channel blocker used for hypertensive patients in Japan, which inhibits sympathetic N-type Ca^{2+} channels in addition to vascular L-type Ca^{2+} channels

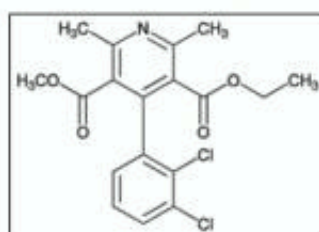


Cardiovascular Therapeutics 27 (2009) 124–139

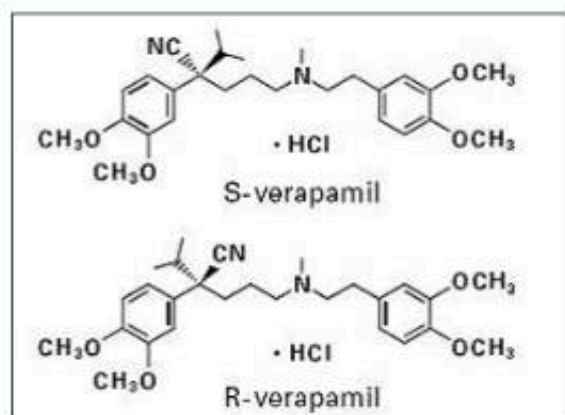
CCBs and Enantiomer



(S)-amlodipine > 1000 fold stronger



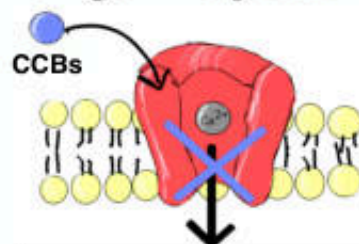
(S)-felodipine > 13 times stronger



(S)-verapamil ~ 20 times more potent than (R)-verapamil with regard to negative dromotropic effects on AV node conduction

Pharmacological Effects and Therapeutic Applications of CCBs

CCBs: Organ System Effects



Smooth muscle

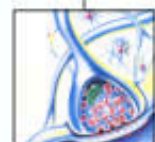
Relaxation of vascular
smooth muscle

* **arterioles** are more
sensitive than veins



Cardiac muscle

- **negative inotropic** effect (cell of myocardium)
- **Impairment** of atrioventricular **conduction** (cells within the His-Perkinje system)



Others

Minimal interference with
stimulus-**secretion** coupling
in **glands and nerve endings**

Cardiovascular Effects of CCBs

Name	Vasodilation (coronary flow)	Suppression of cardiac contractility	Suppression of automaticity (SA node)	Suppression of conductivity (AV node)
Nifedipine	5	1	1	0
Nicardipine	5	0	1	0
Felodipine	5	1	1	0
Amlodipine	5	1	1	0
Diltiazem	3	2	5	4
Verapamil	4	4	5	5

Ranked from no effect (0) to most prominent (5)

Adapted from Goodman & Gilman's: The Pharmacological Basis of Therapeutics 12th Ed

Cardiovascular Effects of CCBs



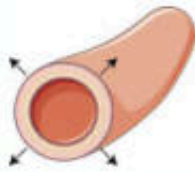
Tachycardia

	Nifedipine	Verapamil	Diltiazem
Vasodilator	++++	++	++
Reflex sympathetic*	+++	+	+
Negative inotropy	++	+++	++
Negative chronotropy	+	+++	+++
Negative dromotropy	+	+++	++

*Reflex sympathetic: tachycardia and positive inotropy

Inotrope: contractility; chronotrope: heart rate; dromotrope: AV-node conduction

The effect of CCBs on SA node and AV node depend on whether the agents delay the recovery of the slow channel



❖ Vascular smooth muscle relaxation

→ vasodilation → ↓ peripheral vascular resistance → ↓ blood pressure

❖ ↓ Cardiac muscle contractility (negative inotropic action)

→ ↓ cardiac output → ↓ blood pressure



❖ ↓ SAN rate (negative chronotropic action)

→ ↓ heart rate (bradycardia) → ↓ cardiac output → ↓ blood pressure

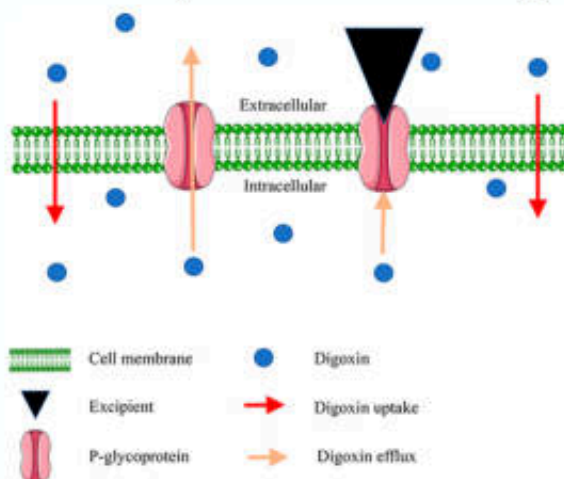
❖ ↓ AVN conduction (negative dromotropic action)

→ termination of supraventricular arrhythmias

Other Effects of Verapamil

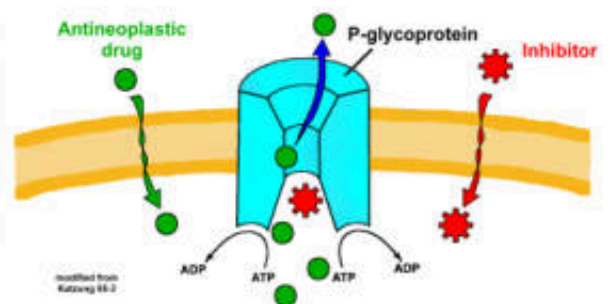
• Blocking the P-glycoprotein (P-gp): **verapamil**

→ decreasing renal clearance of digoxin



Mol. Pharm. 2018;15 (11),4777-5444.

→ partially reversing the resistance of cancer cells to many chemotherapeutic agents



modified from Kufner et al.

creative-bioarray.com

CCBs: Adverse Effects

- ✦ Effects on vascular smooth muscle
- ✦ Myocardial effects
- ✦ Conduction effects
- ✦ Effects on nonvascular smooth muscle

CCBs: Adverse Effects

Effects on vascular smooth muscle:



flushing



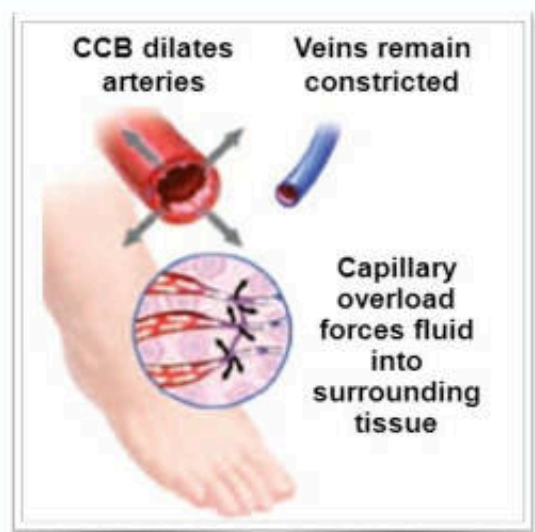
edema



headache



rashes
(cutaneous vasculitis)



CCBs: Adverse Effects

Myocardial effects:



hypotension



heart failure



bradycardia

Conduction effects:



arrhythmia



heart block

Effects on nonvascular smooth muscle:



constipation
(verapamil)



aggravate GI
reflux

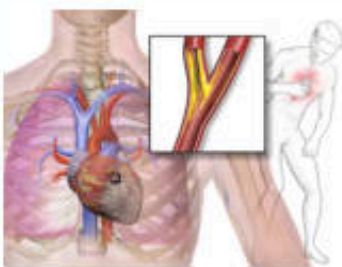


urinary retention



gingival
hyperplasia

CCBs: Therapeutic Uses



Angina



Hypertension



Raynaud's phenomenon



Supraventricular tachycardia
(SVT)



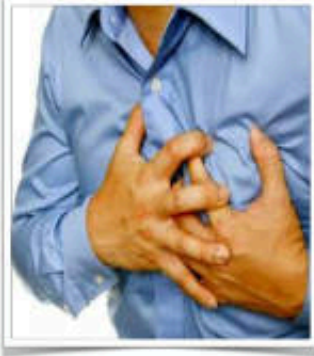
Prophylaxis for
cluster headache



Ischemic neurological
deficit after
subarachnoid hemorrhage

Angina

❖ Angina is chest pain or discomfort caused when your heart muscle doesn't get enough oxygen-rich blood



✦ coronary artery vasodilation

⇒ ↑ O₂ supply

✦ decrease heart rate (HR) and contractility

⇒ ↓ O₂ demand

Hypertension



$$BP = CO \times TPR$$

★ arterial vasodilation

⇒ ↓ TPR

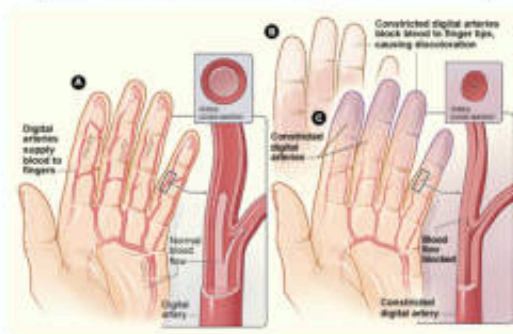
★ reducing the number of electrical impulse and myocardial contractility

⇒ ↓ CO

Raynaud's Phenomenon



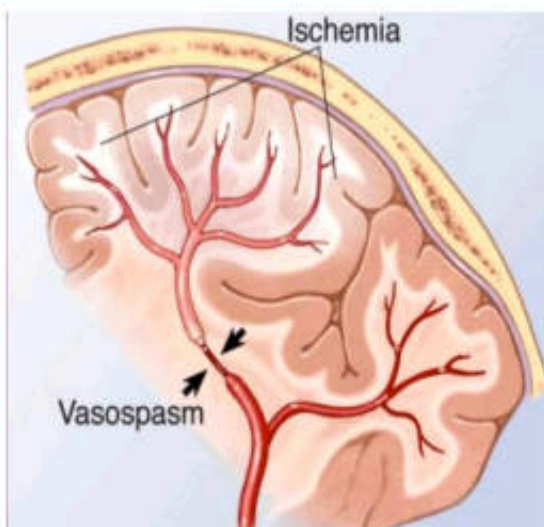
Raynaud phenomenon manifests as recurrent vasospasm of the fingers and toes and usually occurs in response to stress or cold exposure



★ arterial vasodilation

➡ ↑ blood flow to the extremities
(nifedipine is the usual choice)

Ischemic Neurological Deficit After Subarachnoid Hemorrhage



Acute hemorrhage from an aneurysm can physically damage the brain and lead to acute transient global ischemia
: nimodipine

CCBs: Therapeutic Uses

	Nifedipine	Felodipine	Amlodipine	Verapamil	Diltiazem
Hypertension	+	+	+	+	+
Raynaud's phenomenon	+	+			
Angina:					
Stable	+		+	+	+
Prinzmetal's	+		+	+	+
Unstable	+			+	+
Paroxysmal SVT				+	+
Atrial fibrillation, flutter				+	+
Hypertrophic cardiomyopathy				+	
Migraine (prevention)				+	

SVT: Supraventricular tachycardia

